

# Patent Trajectories and M&A Choices

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## Abstract

This paper studies how the evolution of firms’ technologies shapes merger decisions. To observe this evolution, I map firms into a technology space using patent-text embeddings and track both how close firms are and how their relative positions change over time. To identify causality, I use firm exposure to examiner leniency as an instrument for firms’ technological trajectories. I show that mergers are most likely at intermediate levels of technological closeness, consistent with prior evidence that synergies are strongest in this region (Hoberg et al., 2014). Furthermore, conditional on similarity, I find that firms whose technologies are converging more quickly toward an acquirer are less likely to be acquired, suggesting that acquirers avoid targets that are rapidly turning into close competitors, either because such targets gain bargaining power or because such acquisitions raise greater antitrust concerns. To formalize these ideas, I use a simple framework in which merger synergies depend on technological closeness and acquisition frictions depend on how firms’ technologies evolve over time.

## 1 Introduction

Firms continuously face decisions about how to respond to technological change. They can develop new capabilities internally, continue exploring new areas, or acquire technologies from other firms. These choices reflect a fundamental tradeoff between exploiting existing strengths and exploring new opportunities, and they shape both firm boundaries and the allocation of innovation across the economy.

Innovation plays a central role in this process. Patents and research efforts often generate technologies whose use is uncertain and delayed, and many ideas are not immediately commercialized by their original creators (Schankerman, 1998; Serrano, 2010). Instead, firms decide over time whether to develop these technologies internally, continue experimenting, or transfer them across firm boundaries through acquisitions (Bena and Li, 2014). This raises a natural question: how do firms decide which technologies to internalize through M&A and which to leave outside?

Existing evidence shows that technological relatedness matters for these decisions. In corporate

finance, firms are more likely to transact when they have similar product-market language, and these transactions generate stronger gains when similarity creates synergies (Hoberg and Phillips, 2010). In innovation-focused work, technological overlap between acquirers and targets predicts merger pairing and post-merger innovation outcomes (Bena and Li, 2014). More broadly, work in strategy shows that post-acquisition innovation performance depends on how the target’s knowledge base relates to the acquirer’s existing capabilities (Ahuja and Katila, 2001; Cloudt et al., 2006). Related work studies how product-market threats evolve over time, but not how such dynamics shape target selection in M&A (Hoberg et al., 2014; Hoberg and Phillips, 2016).

This paper studies that dynamic margin. I ask whether incumbents respond to technological convergence through acquisition. The key distinction is between firms that are already close and firms that are becoming close. A target that is technologically related to an acquirer may generate synergies if acquired. But a target that is rapidly moving toward the acquirer’s core may also be becoming a stronger future rival. That can raise its bargaining power in a deal and may increase antitrust concerns if the acquisition eliminates an emerging competitor.

To study this problem, I represent firms in a technology space using patent-text embeddings and track both how close firms are and how their positions change over time. This allows me to measure two objects: technological closeness and convergence speed. Closeness captures current relatedness and the potential for merger synergies. Convergence speed captures how quickly one firm’s technologies are moving toward another’s core, and therefore how rapidly a potential target is turning into a closer rival. Because patenting reflects technological positioning before full commercialization, these measures allow me to observe competitive dynamics at an earlier stage than product-market outcomes.

The empirical analysis delivers two main findings. First, acquisitions are most likely at intermediate levels of technological closeness, consistent with prior evidence that synergies are strongest when firms are related but not identical (Hoberg and Phillips, 2010). Second, conditional on closeness, firms whose technologies are converging more quickly toward an acquirer are less likely to be acquired. Thus, incumbents do not appear to buy the firms that are approaching them fastest. Instead, they are more likely to acquire firms that are related yet move more slowly.

To identify causality, I use firm exposure to examiner leniency as an instrument for firms’ technological trajectories. This follows a broader literature that uses quasi-random variation in patent examination and patent grants to study innovation outcomes (Farre-Mensa et al., 2020; Sampat and Williams, 2019). The instrumental-variable estimates show that faster technological convergence reduces the likelihood of acquisition.

To formalize these patterns, I use a simple framework in which merger synergies depend on technological closeness, while acquisition frictions depend on how firms’ technologies evolve over time. The framework captures the idea that some related firms are attractive targets because they generate synergies, while others are unattractive precisely because they are becoming strong independent

competitors.

This paper contributes to three literatures. First, it contributes to the literature on innovation and M&A. Prior work shows that technological overlap shapes merger pairing and post-merger innovation outcomes (Bena and Li, 2014). I show that merger decisions also depend on how firms’ technologies evolve over time.

Second, it contributes to work on text-based measures of similarity and competition. Existing studies use these measures to characterize firm relatedness and competitive pressure (Hoberg and Phillips, 2010; Hoberg et al., 2014; Hoberg and Phillips, 2016). I introduce a dynamic dimension by focusing on how the speed and direction of innovation.

Third, it relates to the literature on innovation and firm scope. I study how the evolution of technological capabilities affects which firms are acquired and which remain independent.

## 2 Data

This section describes the construction of the patent, merger, and examiner-level datasets and the procedures used to link them through firm identifiers.

The analysis begins with the extended patent–firm database of Kogan et al. (2017b), covering all U.S. patents obtained by public firms from 1836–2023. I augment this dataset with Cooperative Patent Classification (CPC) codes, permanent company number PERMCO supplied by the authors. I merge these patents with USPTO PatentView abstracts (1941–2023) and generate 1,024-dimensional PATENTBERT(Lee and Hsiang, 2020) embeddings for each abstract, L2-normalized to lie on the unit sphere. After merging with CRSP–Compustat through the CCM link, the final sample includes 2.6 million firm-linked patents for 7,926 firms during 1961–2023. These embeddings form the basis for dynamic measures of technological proximity and innovation direction. Figure 1 summarizes this distribution.

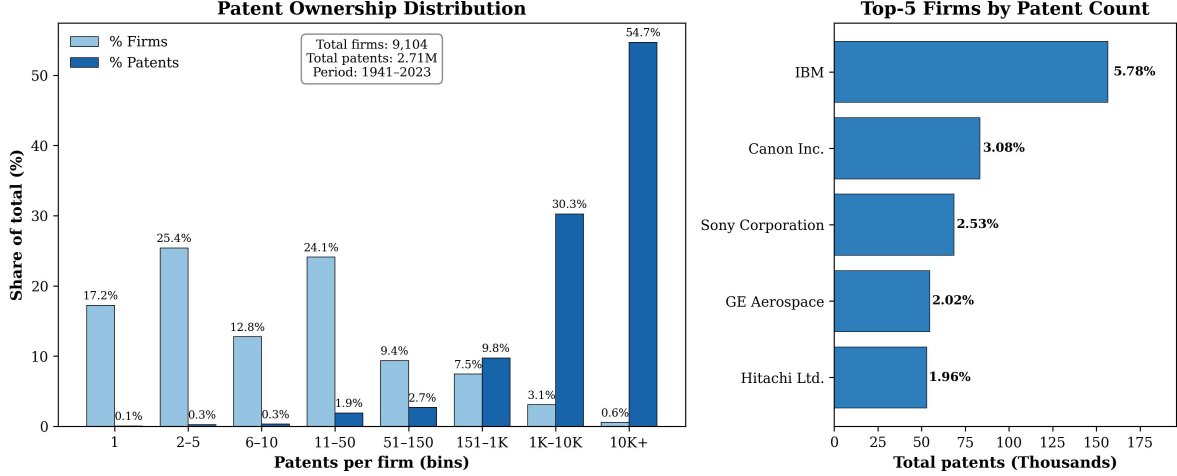


Figure 1: Distribution of historical patent counts for acquirers and targets in the matched public–public sample. The distribution is highly skewed: a small number of firms hold the vast majority of patents, while many firms possess only a limited number of patents.

The global Refinitiv SDC M&A database reports over two million announced transactions world-wide. I restrict the sample to completed deals, yielding 374,547 unique transactions. Because SDC does not provide identifiers consistent with CRSP or Compustat, acquirer and target CUSIPs are matched to CRSP’s monthly security file using six-digit CUSIP roots.

For each CRSP firm, I construct trading intervals between the first and last observed month of activity and define a firm as public if its CUSIP was active on the deal completion date. This procedure identifies 68,621 public acquirers (18.3%), 5,507 public targets (1.5%), and 3,115 transactions in which both sides are public (0.83%).

Merging these CRSP-linked firms with the KPSS patent database through PERMCO shows that 58.1% of acquirers and 36.7% of targets hold at least one patent. The intersection yields 831 public–public transactions where both parties are patent-active. Among these, 455 targets and 144 acquirers hold fewer than ten patents, corresponding to 503 deals in which at least one side exhibits limited patenting activity. This subset forms the baseline for analyzing technological proximity and post-merger innovation outcomes.

To obtain exogenous variation in measured innovative activity, I incorporate the USPTO PatEx dataset (1970–2022). PatEx contains application-level information for more than 13 million filings, including examiner IDs, art-unit assignments, application years, and grant/denial outcomes. I used this data to construct a firm-year level leniency exposure.

### 3 Empirical Design

#### 3.1 Construction of Technological Measures

Each firm–year  $(f, t)$  is represented by a 1,024-dimensional centroid of the embeddings of the abstracts of its patents (PATENTBERT embeddings):

$$\mathbf{c}_{f,t} = \frac{1}{N_{f,t}} \sum_{p \in \mathcal{P}_{f,t}} \mathbf{v}_p.$$

To characterize different horizons of innovation, I construct two additional centroids:

$$\mathbf{c}_{f,t}^{cum} = \frac{1}{N_{f,t}^{cum}} \sum_{\tau \leq t} \sum_{p \in \mathcal{P}_{f,\tau}} \mathbf{v}_p, \quad \mathbf{c}_{f,t}^{(5)} = \frac{1}{N_{f,t}^{(5)}} \sum_{\tau=t-4}^t \sum_{p \in \mathcal{P}_{f,\tau}} \mathbf{v}_p.$$

The cumulative centroid  $\mathbf{c}_{f,t}^{cum}$  captures the firm’s long-run technological base, while the five-year centroid  $\mathbf{c}_{f,t}^{(5)}$  highlights its recent innovation direction. Together, these centroids trace continuous firm trajectories in patent space for all distance and convergence calculations.

**Technological Distance, Convergence, and Threat:** For each acquirer–candidate pair  $(a, j)$  and relative year  $t$ , I compute cosine and Euclidean distances:

$$d_{aj,t}^{cos} = 1 - \frac{\mathbf{c}_{a,t} \cdot \mathbf{c}_{j,t}}{\|\mathbf{c}_{a,t}\| \|\mathbf{c}_{j,t}\|}, \quad d_{aj,t}^{euc} = \|\mathbf{c}_{a,t} - \mathbf{c}_{j,t}\|_2.$$

Smaller values of  $d$  indicate greater similarity in innovation portfolios.

Convergence speed is defined over one- and five-year horizons ( $+S$  denotes convergence):

$$S_{aj,t}^{(1)} = d_{aj,t-1} - d_{aj,t}, \quad S_{aj,t}^{(5)} = \frac{d_{aj,t-5} - d_{aj,t}}{5}.$$

The technological threat index scales convergence by current proximity:

$$T_{aj,t}^{(h)} = \frac{S_{aj,t}^{(h)}}{d_{aj,t} + \kappa}, \quad h \in \{1, 5\}, \quad \kappa = 1.$$

**Pre–Deal Motion in Technology Space:** To summarize the raw dynamics of technological proximity prior to mergers, I track the yearly cosine and Euclidean distances between each realized acquirer–target pair during the ten years preceding the deal. For each pair, I compute the percent change in distance relative to its value ten years before the transaction.

Figure 2 shows the average trajectory. Both distance measures indicate systematic pre–deal convergence: on average, acquirers and targets move closer in technology space as the deal approaches.

This pattern provides descriptive support for the idea that realized mergers occur within the intermediate region where portfolios gradually become more related.

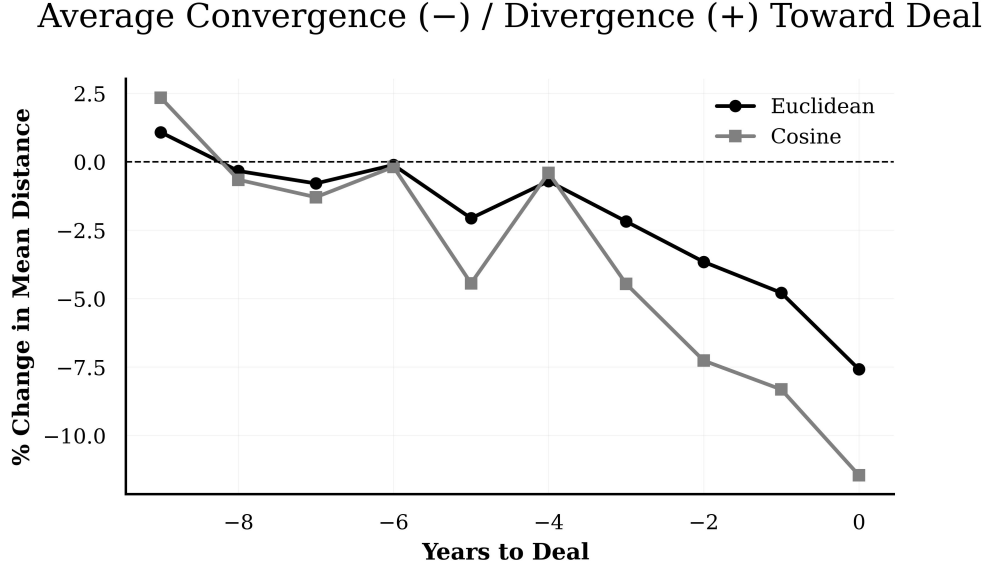


Figure 2: Percent change in cosine and Euclidean distance between acquirer and target over the ten years before the deal. Distances are normalized to zero at  $t = -10$ .

### 3.2 Constructing the Risk Set of Potential Targets

A central empirical challenge is that observing only realized M&A pairs does not reveal which alternative targets were actually feasible in a given year. To estimate target–choice behavior, it is necessary to define the set of potential target firms that an acquirer could plausibly have purchased instead of the observed target.

For each completed public–public deal ( $i \rightarrow j$ ) in year  $t$ , I treat  $i$  as the acquirer and  $j$  as the realized target. I first define the *focal target profile* using  $j$ ’s most recent characteristics measured over the window  $[t - 5, t]$ .

The candidate pool consists of all other (Limiting to 25 candidates per target) patenting firms  $k \neq j$  that:

- (i) operate in the same 2-digit SIC industry during  $[t - 5, t]$ , (ii) are active patenters during the same window, and (iii) satisfy the economic calipers:

$$|\ln A_k - \ln A_j| \leq 1, |\xi_k - \xi_j| \leq 5, |CPC_k - CPC_j| \leq 10.$$

Within this set, all firm characteristics are standardized, and I select the  $K = 25$  nearest neighbors

in Euclidean distance:

$$d_{jk} = \|\tilde{\mathbf{x}}_{j,t} - \tilde{\mathbf{x}}_{k,t}\|_2, \quad k \in \arg \min d_{jk}.$$

These  $K$  firms represent the set of feasible targets available to acquirer  $i$  in year  $t$ . The realized target  $j$  is then compared to its feasible alternatives on technological distance, convergence speed, threat, and financial covariates. This construction allows estimation of target-choice models that respect realistic substitution patterns and acquirer-specific opportunity sets.

Table 1 reports balance between realized targets and their matched candidate firms across standard financial and innovation variables. The lack of systematic differences supports the empirical validity of the constructed risk sets.

Table 1: Balance diagnostics for target-candidate matching

| Variable                          | Target mean | Candidate-Target mean | Difference | t-stat | p-value |
|-----------------------------------|-------------|-----------------------|------------|--------|---------|
| $\ln AT$                          | 5.907       | 5.136                 | -0.771     | -0.485 | 0.630   |
| ROA                               | -0.035      | -0.028                | 0.007      | 0.196  | 0.845   |
| $Q$                               | 1.134       | 0.963                 | -0.172     | 0.566  | 0.576   |
| Leverage                          | 0.226       | 0.141                 | -0.086     | -1.087 | 0.282   |
| Cash/AT                           | 0.216       | 0.195                 | -0.021     | 0.117  | 0.907   |
| R&D/AT                            | 0.108       | 0.080                 | -0.027     | -0.099 | 0.922   |
| CAPX/AT                           | 0.038       | 0.035                 | -0.004     | 1.939  | 0.058   |
| MKVALT (\$M)                      | 4,156       | 3,660                 | -496       | 1.048  | 0.305   |
| $\xi_{\text{mean}}^{\text{real}}$ | 6.408       | 5.610                 | -0.798     | -0.341 | 0.739   |
| Citations <sub>mean</sub>         | 18.584      | 15.642                | -2.942     | 0.059  | 0.954   |
| CPC unique count                  | 16.000      | 33.676                | 17.676     | -0.873 | 0.402   |

*Notes.* This table compares realized **targets** to their matched feasible candidates within a five-year lookback window. Matching restricts to firms in the same two-digit SIC and with nonzero patenting activity. Candidates are chosen using standardized Euclidean distance over firm characteristics  $\{\ln AT, ROA, Q, LEV, CASH/AT, XRD/AT, CAPX/AT, MKVALT, \xi_{\text{mean}}^{\text{real}}, \text{Citations}_{\text{mean}}, \text{CPC unique count}\}$ . Differences represent candidate minus target means across deals; the absence of significant differences indicates strong technological comparability.  $\xi$  represents the KPSS market response to each patent.

### 3.3 Baseline Target-Choice Results (OLS)

Using the acquirer-year risk sets, I estimate the probability that candidate  $k$  becomes the chosen target in deal-year  $t$  as a function of its technological threat to the acquirer. The baseline specification is:

$$is\_target_{ikt} = \alpha_i + \gamma_t + \beta T5\_euc, ikt + X'_{ikt} \delta + \varepsilon_{ikt}.$$

where  $\alpha_i$  denotes acquirer fixed effects,  $\gamma_t$  denotes deal-year fixed effects,  $T5\_euc$  is the five-year technological threat index, and  $X_{ikt}$  includes firm-level covariates such as changes in assets, ROA,  $q$ , leverage, cash holdings, R&D intensity, capital investment, market value, and innovation activity.

Table 2: Baseline OLS: Threat and Target Selection

|                             | Coefficient           | Std. Error           | <i>t</i> -stat |
|-----------------------------|-----------------------|----------------------|----------------|
| $T5_{euc}$                  | -0.174                | 0.040                | -4.38          |
| $\Delta \ln(\text{Assets})$ | 0.0013                | 0.0010               | 1.38           |
| $\Delta \text{ROA}$         | 0.0001                | 0.0013               | 0.12           |
| $\Delta q$                  | 0.00009               | 0.00010              | 0.96           |
| $\Delta \text{Leverage}$    | 0.0018                | 0.0027               | 0.68           |
| $\Delta \text{Cash/AT}$     | 0.0016                | 0.0021               | 0.74           |
| $\Delta \text{R\&D/AT}$     | -0.0010               | 0.0030               | -0.34          |
| $\Delta \text{CAPX/AT}$     | -0.0011               | 0.0080               | -0.14          |
| $\Delta \text{MktVal}$      | $2.8 \times 10^{-9}$  | $4.1 \times 10^{-9}$ | 0.68           |
| $\Delta \text{Patents}$     | $1.7 \times 10^{-6}$  | $1.3 \times 10^{-6}$ | 1.23           |
| $\Delta \text{Citations}$   | $-3.9 \times 10^{-8}$ | $3.5 \times 10^{-8}$ | -1.13          |
| Acquirer FE ( $\alpha_i$ )  | Yes                   |                      |                |
| Deal-year FE ( $\gamma_t$ ) | Yes                   |                      |                |
| Clusters                    | 255                   |                      |                |
| Observations                | 18,513                |                      |                |

*Notes:* This table reports OLS estimates from the target-choice regression using acquirer-year risk sets. The dependent variable equals one if candidate  $k$  is the realized target in deal-year  $t$ .  $T5_{euc}$  is the five-year technological threat index. All specifications include acquirer fixed effects and deal-year fixed effects. Standard errors are clustered at the acquirer level.

The OLS estimate shows a negative and statistically significant association between threat and target choice: a one-standard-deviation increase in  $T5_{euc}$  ( $SD = 0.018$ ) lowers the probability of being selected as the target by roughly 0.32 percentage points.

Economically, firms are *less likely* to acquire candidates that are rapidly converging toward their technological core. This pattern suggests defensive avoidance rather than aggressive consolidation. Since  $T5_{euc}$  is measured with error due to patent-grant randomness and imperfect distance measurement, attenuation bias likely pushes the OLS coefficient toward zero; the true effect is expected to be stronger.

The measured threat index  $T5_{euc}$  depends on observed patent grants, which are not exogenous. Firms anticipating a merger may shift their patenting toward technologies that appear attractive to a potential acquirer, creating reverse causality. Common shocks like industry-level technology cycles or funding conditions can simultaneously affect patenting intensity and the likelihood of M&A activity, generating omitted-variable bias. In addition, random noise in patent embeddings introduces measurement error that attenuates the OLS coefficient toward zero. Together, these issues can bias the estimated relationship between technological threat and target choice, potentially

masking the true negative effect. To recover exogenous variation in perceived innovation intensity, I use examiner leniency exposure as an instrument, which affects patent grant outcomes but is orthogonal to firms’ acquisition intent.

### 3.4 Identification Strategy—Instrumental Variable

Examiner-specific heterogeneity in granting behavior provides plausibly exogenous variation in patent approvals, following [Frakes and Wasserman \(2014, 2020\)](#), [Galasso and Schankerman \(2018\)](#), and [Sampat and Williams \(2019\)](#). I construct examiner leniency using the USPTO Patent Examination Research Dataset (PatEx, 2022 release), which covers more than 13 million publicly visible provisional and non-provisional U.S. applications and over one million PCT filings. PatEx is compiled from the Patent Examination Data System (PEDS) as of June 2023 and includes examiner identifiers, art units, filing dates, disposition outcomes, and status codes.

I apply the following filters before constructing the instrument. First, I restrict to applications with (i) a named examiner, (ii) a valid art unit, (iii) a non-missing filing date, and (iv) an issued U.S. patent number. I then match issued applications to the [Kogan et al. \(2017b\)](#) patent panel by patent number in order to attach a CRSP firm identifier (`permco`). Second, I collapse examiners to an “*examiner key*,” defined as the examiner’s full name interacted with the art unit. I drop examiner keys with fewer than five observed applications in a given year to avoid high-variance examiners. Third, I drop art units with fewer than twenty observed applications and remove art units whose grant rate is essentially degenerate (grant ratio  $\leq 1\%$  or  $\geq 99\%$ ). Figure 3 shows the resulting distribution of art-unit size and grant rates.

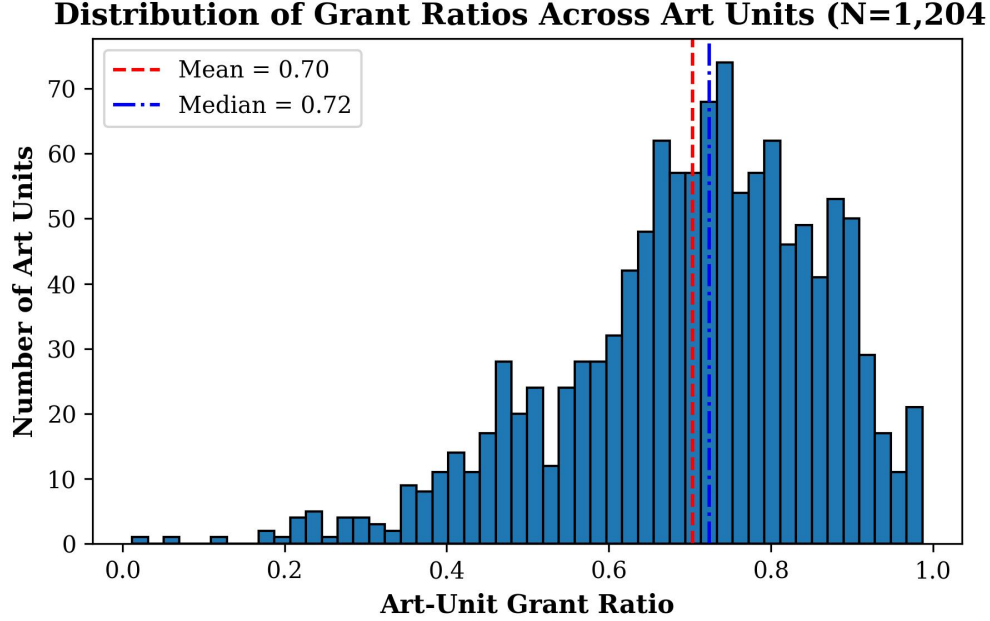


Figure 3: Art-unit level application grant rates after filtering on examiner coverage and art-unit activity.

For each application  $a$  examined by examiner  $e$  in year  $t$ , I compute (i) the examiner's leave-one-out grant rate and (ii) its residual within art-unit-year:

$$Z_{aet} = \frac{G_{e,t} - \text{Grant}_a}{N_{e,t} - 1}, \quad Z_{aet}^{\text{resid}} = Z_{aet} - \bar{Z}_{aut},$$

where  $N_{e,t}$  and  $G_{e,t}$  are the total examined and granted applications for examiner  $e$  in year  $t$ ,  $\text{Grant}_a$  is the grant outcome for application  $a$ , and  $\bar{Z}_{aut}$  is the mean leniency in art unit  $au$  and year  $t$ . The first term removes mechanical dependence on the focal application; the second removes art-unit-year shifts in examination policy.

Finally, I aggregate these examiner-level shocks to the firm-year level. Let  $\mathcal{A}_{f,t}$  be the set of granted applications filed in year  $t$  by firm  $f$  (matched via `permno`). The firm's exposure to examiner leniency is:

$$Z_{f,t}^{\text{firm}} = \frac{1}{|\mathcal{A}_{f,t}|} \sum_{a \in \mathcal{A}_{f,t}} Z_{aet}, \quad Z_{f,t}^{\text{resid}} = \frac{1}{|\mathcal{A}_{f,t}|} \sum_{a \in \mathcal{A}_{f,t}} Z_{aet}^{\text{resid}}.$$

This produces a panel of 8,191 CRSP-linked firms over 1970–2022 (53 years), with 2,268,139 granted applications contributing to the instrument. Figure 4 shows the cross-sectional distribution of  $Z_{f,t}^{\text{resid}}$  across firm-years. These measures provide firm-year shocks to the probability of patent grant, which I later use to construct instruments for patenting success and firm dynamics in the M&A analysis.

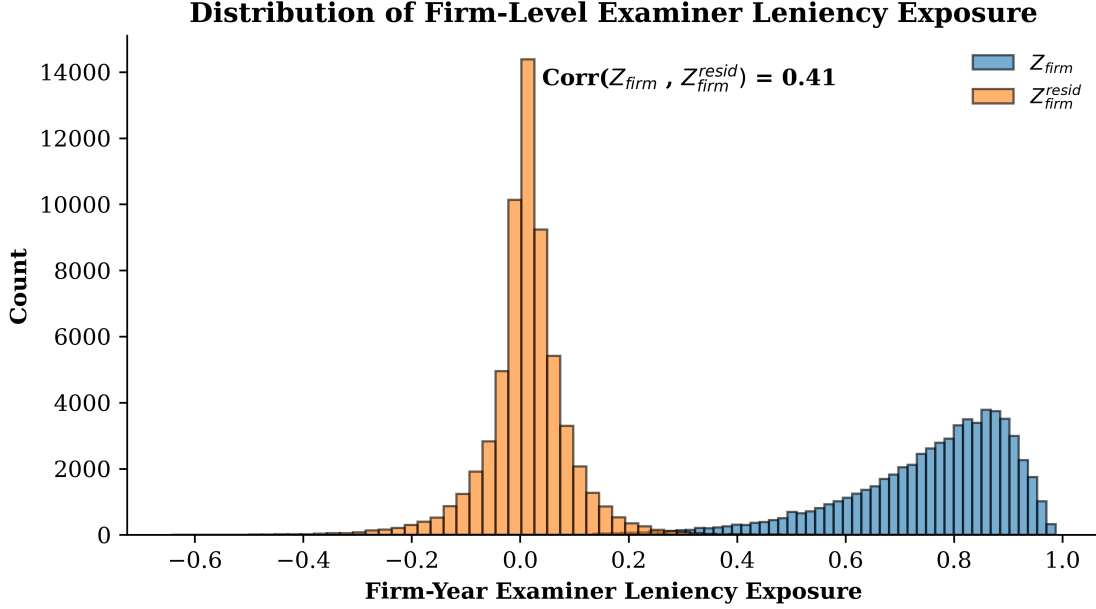


Figure 4: Distribution of firm-year exposure to examiner leniency,  $Z_{f,t}$ , and its residualized counterpart,  $Z_{f,t}^{\text{resid}}$ . Both measures reflect variation in examiner grant propensity across firms and years, with  $Z_{f,t}^{\text{resid}}$  purging art-unit-year effects.

### 3.5 Instrumental Variable Results

This section reports the two-stage least squares estimates using examiner–leniency exposure as an instrument for the technological threat index  $T5_{euc}$ . As described earlier, examiner assignment within art units is quasi-random, and examiners differ persistently in grant propensity. Firms exposed to more lenient examiners appear more technologically active not because they strategically accelerate innovation, but because a greater share of their applications are approved. This quasi-random variation generates exogenous shifts in measured convergence and threat that are orthogonal to acquisition intent.

#### 3.5.1 First Stage: Leniency and Perceived Threat

The first-stage regression relates the threat index  $T5_{euc}$  for acquirer–candidate pair  $(i, k)$  in year  $t$  to the five-year residualized examiner–leniency exposure:

$$T5_{euc,ikt} = \pi_1 Z5_{ikt}^{\text{resid}} + X'_{ikt} \rho + \mu_i + \tau_t + \nu_{ikt}.$$

where  $\mu_i$  are acquirer fixed effects,  $\tau_t$  are deal-year fixed effects, and  $X_{ikt}$  contains changes in firm-level characteristics.

Table 3 reports the results. The coefficient on  $Z5^{\text{resid}}$  is positive and statistically significant: more lenient examiner exposure increases the measured technological threat between an acquirer and a

candidate. This reflects the fact that when a candidate’s patents are more likely to be granted for reasons unrelated to strategy, its recent patent trajectory shifts inward toward the acquirer in embedding space. The Kleibergen–Paap statistic of 11.93 indicates a relevant first stage.

Table 3: First Stage: Examiner Leniency and Perceived Threat

|                             | Coefficient           | Std. Error            | <i>t</i> -stat |
|-----------------------------|-----------------------|-----------------------|----------------|
| $Z5^{resid}$                | 0.0100                | 0.0029                | 3.45           |
| $\Delta \ln(\text{Assets})$ | −0.00135              | 0.00033               | −4.09          |
| $\Delta \text{ROA}$         | −0.00015              | 0.00081               | −0.19          |
| $\Delta q$                  | −0.00013              | 0.00008               | −1.55          |
| $\Delta \text{Leverage}$    | 0.00035               | 0.00135               | 0.26           |
| $\Delta \text{Cash/AT}$     | 0.00000               | 0.00093               | 0.00           |
| $\Delta \text{R\&D/AT}$     | −0.00294              | 0.00170               | −1.73          |
| $\Delta \text{CAPX/AT}$     | 0.00190               | 0.00401               | 0.47           |
| $\Delta \text{MktVal}$      | $3.4 \times 10^{-9}$  | $1.3 \times 10^{-9}$  | 2.56           |
| $\Delta \text{Patents}$     | $1.26 \times 10^{-6}$ | $2.94 \times 10^{-7}$ | 4.27           |
| $\Delta \text{Citations}$   | $1.2 \times 10^{-9}$  | $1.8 \times 10^{-8}$  | 0.07           |
| Acquirer FE                 | Yes                   |                       |                |
| Deal-Year FE                | Yes                   |                       |                |
| Kleibergen–Paap $F$         | 11.93                 |                       |                |
| Clusters                    | 239                   |                       |                |
| Observations                | 15,634                |                       |                |

*Notes:* The dependent variable is the five-year technological threat index  $T5_{euc}$ . The instrument is the five-year residualized examiner-leniency exposure  $Z5^{resid}$ . Standard errors cluster at the acquirer level.

### 3.5.2 Second Stage: Threat and Target Choice

The second stage estimates the causal effect of technological threat on the probability that a candidate is chosen as the target:

$$is\_target_{ikt} = \alpha_i + \gamma_t + \beta \widehat{T5_{euc,ikt}} + X'_{ikt} \delta + \varepsilon_{ikt}.$$

Table 4 reports the results. The IV estimate of  $\beta$  is negative and statistically significant: a one-standard-deviation increase in the threat index ( $SD = 0.018$ ) decreases the probability of being selected by approximately 3.6 percentage points. This effect is substantially larger in magnitude than the corresponding OLS estimate, consistent with attenuation from measurement error in observed patent signals.

Table 4: Second Stage: Threat and the Probability of Being Acquired

|                             | Coefficient            | Std. Error            | <i>t</i> -stat |
|-----------------------------|------------------------|-----------------------|----------------|
| $\widehat{T5_{euc}}$        | -1.953                 | 0.854                 | -2.29          |
| $\Delta \ln(\text{Assets})$ | -0.00040               | 0.00179               | -0.22          |
| $\Delta \text{ROA}$         | -0.00030               | 0.00213               | -0.14          |
| $\Delta q$                  | -0.00008               | 0.00026               | -0.30          |
| $\Delta \text{Leverage}$    | 0.00361                | 0.00417               | 0.87           |
| $\Delta \text{Cash/AT}$     | 0.00127                | 0.00329               | 0.38           |
| $\Delta \text{R\&D/AT}$     | -0.00604               | 0.00568               | -1.06          |
| $\Delta \text{CAPX/AT}$     | 0.00201                | 0.01344               | 0.15           |
| $\Delta \text{MktVal}$      | $7.0 \times 10^{-9}$   | $5.4 \times 10^{-9}$  | 1.29           |
| $\Delta \text{Patents}$     | $4.05 \times 10^{-6}$  | $1.91 \times 10^{-6}$ | 2.12           |
| $\Delta \text{Citations}$   | $-6.96 \times 10^{-8}$ | $6.31 \times 10^{-8}$ | -1.10          |
| Acquirer FE                 | Yes                    |                       |                |
| Deal-Year FE                | Yes                    |                       |                |
| Clusters                    | 239                    |                       |                |
| Observations                | 15,634                 |                       |                |

*Notes:* The dependent variable equals one if candidate  $k$  is the realized target for acquirer  $i$  in year  $t$ . The threat index is instrumented using examiner-leniency exposure. Standard errors cluster at the acquirer level.

### 3.6 Empirical Patterns in Technological Motion

The threat index  $T5_{euc}$  can be positive or negative because it reflects the ratio of convergence speed to current technological distance. A firm may be close but drifting away (small  $d$  and negative  $S$ ), or distant but moving sharply toward the incumbent (large  $d$  but strongly positive  $S$ ). The two panels in Figure 5 illustrate this geometry. When the technological gap is large, even a moderate directional shift creates a “big jump” in measured convergence. Conversely, for firms already close to the incumbent’s core, small portfolio adjustments generate only a “small jump.” This variation produces both positive and negative values of  $T5_{euc}$  and provides the identifying variation needed for the empirical tests.

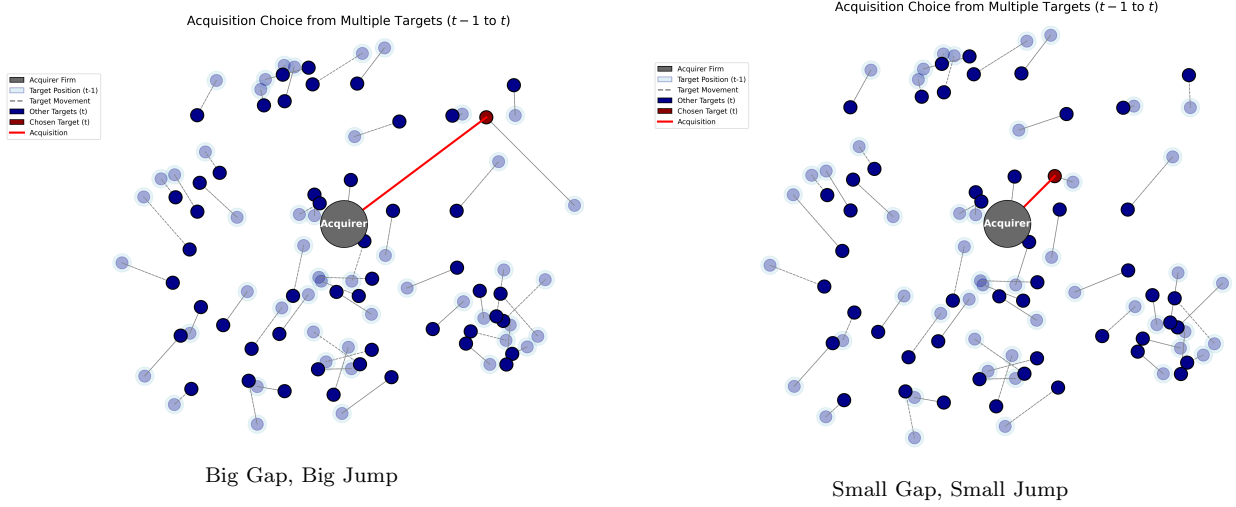


Figure 5: Big Gap, Big Jump (left) and Small Gap, Small Jump (right). The figure is to visually illustrate the difference, and it is not from the actual deals.

The IV estimates in Table 5 show that these geometric patterns have economically meaningful consequences. First, the positive coefficient on technological distance confirms that acquirers favor candidates in the absorbable band: firms that are neither too close nor too distant. This is consistent with the model's hump-shaped surplus in  $S(d)$ , in which intermediate distances yield the strongest complementarities. Second, the strongly negative coefficient on convergence speed indicates that firms moving rapidly toward the incumbent's core are systematically avoided. This aligns with the collision-zone logic of the framework: high  $s$  raises the target's bargaining power and expected regulatory burden, lowering  $V_I(d, s)$  and making the acquisition unattractive.

Table 5: Instrumental Variable Results: Complementarity vs. Speed

|                                              | First Stage |          | Second Stage (2SLS)  |                                                            |
|----------------------------------------------|-------------|----------|----------------------|------------------------------------------------------------|
|                                              | KP $F$      | $p$ -val | IV Coef. ( $\beta$ ) | Effect of $+1\sigma$ in regressor                          |
| <b>(1) Complementarity:</b> $d_{\text{euc}}$ | 12.08       | 0.001    | +0.192**             | $1\sigma \uparrow \Rightarrow +2.1 \text{ pp } \uparrow$   |
| <b>(2) Speed:</b> $S5_{\text{euc}}$          | 11.32       | 0.001    | -1.363**             | $1\sigma \uparrow \Rightarrow -3.4 \text{ pp } \downarrow$ |
| Obs.                                         | 26,765      |          | 26,765 / 16,424      |                                                            |
| Clusters (acq_permco)                        | 244         |          | 244 / 239            |                                                            |

*Notes:* This table reports instrumental-variable estimates of technological complementarity ( $d_{\text{euc}}$ ) and convergence speed ( $S5_{\text{euc}}$ ) on target choice. The first-stage columns show Kleibergen–Paap  $F$ -statistics and  $p$ -values for the examiner-leniency instruments. The second-stage columns report IV coefficients and the implied marginal effects of a one-standard-deviation increase in the regressor on the probability of being acquired. Standard errors are clustered at the acquirer level.

Table 6 provides a complementary test using within-risk-set differences. Conditional on being feasible candidates in the same acquirer-year, realized targets do not differ meaningfully in static

distance or overall threat level, but they exhibit significantly lower convergence speed. This pattern confirms that avoidance of fast-approaching rivals holds even after conditioning on observables that define the feasible set. Although acquirers and targets tend to converge technologically before the deal, the chosen targets are those whose trajectories remain relatively slower and more absorbable.

Table 6: Targets vs. Feasible Candidates: Technological Characteristics

| Variable                                         | Mean       | Std. Dev. |
|--------------------------------------------------|------------|-----------|
| $\Delta T5_{\text{euc}}$ (Threat Index)          | -0.0013    | 0.0220    |
| $\Delta S5_{\text{euc}}$ (Convergence Speed)     | -0.0026*** | 0.0077    |
| $\Delta d_{\text{euc}}$ (Technological Distance) | 0.0055     | 0.1114    |

*Notes:* Differences are computed within each acquirer-year risk set as (Target – mean of candidate pool). *t*-tests are for  $H_0$ : mean difference = 0.  
\*\*\* $p < 0.01$ .

Taken together, the evidence indicates that negative values of the threat index correspond to dynamics favorable for acquisition, firms drifting away or converging only gradually, while positive and high values capture the rapid motion associated with the collision zone. The empirical patterns mirror the theoretical predictions: incumbents acquire complementary, slow-moving targets within the absorbable band and systematically avoid candidates whose innovation paths are bending too quickly toward their technological core.

### 3.7 Interpretations and Connection to the Theoretical Framework

The IV results align closely with the predictions of the simple framework developed in the theory section. In the model, merger surplus is hump-shaped in technological distance and decreasing in convergence speed once the candidate enters the collision zone: for a given distance in the absorbable band ( $d_L, d_H$ ), higher  $s$  raises the target’s bargaining power and expected regulatory costs, reducing the incumbent’s net gain  $V_I(d, s)$ . The empirical threat index  $T5_{\text{euc}}$  is a direct analogue of this geometry—capturing convergence speed scaled by distance—and the examiner-leniency instrument generates exogenous shifts in the observable component of this motion.

**Integration frictions and absorptive capacity.** The positive IV effect for technological complementarity is consistent with mergers occurring in the absorbable region of the model. At intermediate distances, firms’ knowledge bases differ enough to generate synergies but not so much that integration becomes infeasible. Candidates that are too distant contribute little new capability, while candidates that are too close offer little incremental synergy. The empirical fact that realized targets cluster in this middle region matches the hump-shaped synergy term  $S(d)$ .

**Bargaining power and dynamic rivalry.** The strong negative causal effect of convergence speed reflects the mechanism by which high  $s$  enhances the target’s bargaining weight  $\beta(d, s)$ . A

fast-approaching candidate represents a future competitive threat, increasing its outside option and reducing the incumbent’s share of surplus. The IV estimates show that when a quasi-random shock makes a candidate *appear* to be converging more rapidly, the incumbent becomes significantly less likely to acquire it. This is precisely the comparative static implied by the collision-zone threshold  $\bar{s}(d)$ .

**Regulatory scrutiny in the collision zone.** Rapid convergence also amplifies expected regulatory costs. When portfolios grow too similar, consolidation concentrates innovation capacity, raising the likelihood of antitrust review. In the model, this is captured by  $R(d)$  increasing as distance shrinks. Empirically, the fact that convergence speed has a larger causal effect than distance suggests that acquirers respond not only to static similarity but to the rate at which a transaction would move the pair into a regulatory-sensitive region.

**Information frictions and perceived trajectory risk.** The IV results highlight a perception channel. Examiner-leniency shocks inflate observed innovation motion by increasing the likelihood that marginal patents are granted. This makes the candidate *look* more convergent even if its underlying R&D direction is unchanged. Because incumbents base acquisition choices on observed  $(d, s)$  rather than latent fundamentals, measurement error in  $T5_{euc}$  attenuates OLS, whereas the IV estimates recover the causal avoidance response predicted by the model.

**Unified mechanism.** Taken together, the evidence supports a mechanism in which incumbents partition potential targets into three geometric regions. Firms that are too distant generate little synergy; fast-approaching or overly similar firms fall into the collision zone because of bargaining power and regulatory frictions; and acquisitions occur primarily among candidates in the absorbable band—those at intermediate distances with moderate or slow convergence. The strengthened IV coefficient provides a causal validation of the model’s core prediction: dynamic innovation motion, not static technological overlap, determines which firms incumbents choose to acquire and which they prefer to leave as future competitors.

## 4 Simplified Conceptual Framework

This section provides a minimal framework to organize the empirical analysis. The goal is not to solve a full model, but to fix ideas and clarify how technological distance and technological motion shape merger decisions.

### 4.1 Technology Space and State Variables

Each firm is represented as a point in a  $K$ -dimensional technology space. Let

$$c_I(t), c_J(t) \in \mathbb{R}^K$$

denote the centroids of the patent portfolios of the incumbent  $I$  and the candidate  $J$  at time  $t$ . The relative position of the candidate is

$$x(t) \equiv c_J(t) - c_I(t), \quad d(t) \equiv \|x(t)\|.$$

The scalar  $d(t)$  is the technological distance between the two firms.

What matters for merger decisions is not only where firms are located, but also how this distance changes over time. Define convergence speed as

$$s(t) \equiv -\frac{d}{dt}d(t).$$

A positive value  $s(t) > 0$  means that the candidate is moving toward the incumbent, while  $s(t) < 0$  means it is moving away.

The pair  $(d, s)$  summarizes the state relevant for acquisition decisions. Distance captures current similarity, while speed captures how this similarity evolves.

## 4.2 Merger Surplus

An acquisition is economically meaningful only when firms are sufficiently related. If they are too distant, their technologies do not combine easily. If they are too close, combining them adds little value. Let total merger surplus be

$$S(d),$$

with

$$S(d) > 0 \quad \text{only for } d \in (d_L, d_H),$$

where  $0 < d_L < d_H$  define the lower and upper bounds of the *absorbable region*. Outside this region, merger surplus is non-positive.

For intuition, assume

$$S(d) = \alpha(d - d_L)(d_H - d), \quad \alpha > 0,$$

so that surplus is maximized at an intermediate distance.

## 4.3 Bargaining

Even when merger surplus is positive, an acquisition may not occur because the incumbent does not capture the full surplus from the deal.

Let  $\beta(d, s) \in [0, 1]$  denote the share of surplus received by the candidate, so the incumbent receives  $1 - \beta(d, s)$ . I assume

$$\beta(d, s) = \beta_0 + \beta_s s,$$

where  $\beta_0 \in [0, 1]$  is the baseline bargaining weight of the candidate and  $\beta_s > 0$  captures how

bargaining power increases with convergence speed. The idea is that a candidate that is rapidly approaching the incumbent becomes a stronger standalone competitor, which improves its outside option and raises its share of surplus.

#### 4.4 Regulatory and Integration Costs

Acquisitions may also face regulatory scrutiny. Let

$$R(d, s) = r_0 + r_1(d_{\text{crit}} - d) + r_2 \max\{s, 0\},$$

where  $r_0 \geq 0$  is a baseline regulatory cost,  $r_1 > 0$  captures how scrutiny increases as firms become more similar,  $d_{\text{crit}}$  is the distance below which firms are considered close substitutes, and  $r_2 \geq 0$  captures how regulatory costs increase when the candidate is rapidly converging.

In addition, let  $K \geq 0$  denote a fixed integration cost.

#### 4.5 Acquisition Decision

The incumbent's net gain from acquiring the candidate is

$$V_I(d, s) = (1 - \beta(d, s))S(d) - R(d, s) - K.$$

The incumbent acquires the candidate if and only if

$$V_I(d, s) \geq 0.$$

Substituting the functional forms gives

$$V_I(d, s) = (1 - \beta_0 - \beta_s s)S(d) - r_0 - r_1(d_{\text{crit}} - d) - r_2 \max\{s, 0\} - K.$$

For  $s > 0$  and  $S(d) > 0$ , we have

$$\frac{\partial V_I(d, s)}{\partial s} = -\beta_s S(d) - r_2 < 0,$$

so faster convergence reduces the incumbent's net gain from acquisition.

#### 4.6 Speed Cutoff

Because  $V_I(d, s)$  is decreasing in  $s$  when  $S(d) > 0$ , there exists a cutoff  $\bar{s}(d)$  such that the incumbent acquires if

$$s \leq \bar{s}(d),$$

and does not acquire otherwise. Solving for the cutoff yields

$$\bar{s}(d) = \frac{(1 - \beta_0)S(d) - r_0 - r_1(d_{\text{crit}} - d) - K}{\beta_s S(d) + r_2}.$$

The numerator represents the static net gain from acquisition, while the denominator captures the dynamic costs associated with convergence.

## 4.7 Regions and Predictions

Define the acquisition region

$$\mathcal{A} = \{(d, s) : d \in (d_L, d_H), s \leq \bar{s}(d)\},$$

and the collision region

$$\mathcal{C} = \{(d, s) : d \in (d_L, d_H), s > \bar{s}(d)\}.$$

Firms in  $\mathcal{A}$  are sufficiently related and move slowly enough to be acquired. Firms in  $\mathcal{C}$  are related but approach too quickly, making acquisition unattractive.

The framework delivers three predictions:

1. Acquisition likelihood is highest at intermediate distances.
2. Conditional on distance, acquisition likelihood declines with convergence speed.
3. Stronger bargaining sensitivity ( $\beta_s$ ) or regulatory pressure ( $r_2$ ) reduces the set of firms that are acquired.

## 4.8 Link to the Data

In the empirical analysis, technological distance  $d$  is measured using the distance between firm-level patent embeddings. Convergence speed  $s$  is measured as the change in distance over time. The acquisition rule motivates the empirical design, which compares realized targets to feasible alternatives and examines how acquisition probabilities vary with  $d$  and  $s$ . The examiner-leniency instrument provides exogenous variation in technological trajectories.

## 5 Conclusion

This paper shows that the dynamics of firms' technological portfolios play an important role in selection into mergers. By representing firms as evolving trajectories in a patent based technology space, the analysis separates two primitives of innovation motion: the technological distance between firms at a given moment and the speed at which a candidate's research direction moves toward or away from an incumbent. A simple framework illustrates why incumbents value candidates

at intermediate distances but avoid those whose portfolios converge too quickly. Fast convergence strengthens the target’s outside option and increases expected regulatory scrutiny, creating a collision region where acquisition becomes unattractive even when firms are technologically related.

Empirically, transformer based patent embeddings make it possible to trace these trajectories at the firm year level. Acquirer target pairs lie in an interior band of technological distances and exhibit gradual pre deal convergence. Conditional on distance, the likelihood that a candidate is acquired declines sharply with convergence speed. This pattern becomes stronger when I instrument for the evolution of granted patent portfolios using examiner driven variation in grant propensity. The evidence indicates that dynamic innovation motion, not only static technological similarity, is a first order determinant of which firms are absorbed and which remain independent competitors.

Overall, the results suggest that merger analysis should account for both the location of firms in technology space and the speed at which they move through it. Dynamic technological motion contains economically meaningful information about future rivalry, bargaining, and regulatory exposure, and these forces help shape the structure of acquisition flows. Future research can build on this perspective by studying how technological trajectories interact with acquirer heterogeneity, post merger integration, and industry level innovation cycles.

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## Appendix

Table 7 summarizes the variables I have used for the regressions:

Table 7: Summary of Variables in the Firm-Year Panel

| Variable                                     | Definition and Construction                                                |
|----------------------------------------------|----------------------------------------------------------------------------|
| <i>Identifiers and Timing</i>                |                                                                            |
| permco, gvkey                                | Firm identifiers (CRSP and Compustat linkage)                              |
| filing_year                                  | Patent application filing year; used as panel year index                   |
| cal_year                                     | Calendar-aligned fiscal year (based on fyr)                                |
| <i>Patent and Innovation Measures (KPSS)</i> |                                                                            |
| n_patents_window                             | Number of granted patents in a 10-year rolling window                      |
| centroid_1024, radius_cos                    | Patent embedding centroid and portfolio dispersion (technological breadth) |
| cites10_sum, cites10_mean                    | Forward citations within 10 years of grant (sum and mean)                  |
| xi10_sum, xi10_mean                          | Innovation impact index (sum and mean)                                     |
| cpc10_unique                                 | Count of unique CPC subclasses in 10-year window                           |
| <i>Examiner-Based Instrument</i>             |                                                                            |
| Z                                            | Examiner leniency (leave-one-out grant rate) averaged at firm-year level   |
| Z_resid                                      | Residualized leniency (within art-unit $\times$ year)                      |
| n_grants                                     | Number of granted applications used in instrument computation              |
| <i>Financial Variables (Compustat)</i>       |                                                                            |
| at, sale, ib                                 | Total assets, net sales, and income before extraordinary items             |
| roa                                          | Return on assets: $ROA = \frac{IB}{AT}$                                    |
| lev                                          | Leverage ratio: $LEV = \frac{DLTT+DLC}{AT}$                                |
| cash_at                                      | Cash holdings ratio: $\frac{CHE}{AT}$                                      |
| xrd_at                                       | R&D intensity: $\frac{XRD}{AT}$                                            |
| capx_at                                      | Investment intensity: $\frac{CAPX}{AT}$                                    |
| wc_at                                        | Working capital ratio: $\frac{WCAP}{AT}$                                   |
| q                                            | Tobin's $Q = \frac{MKV+ALT+PST+KRV-CEQ+DT}{AT}$                            |
| <i>Dynamic and Log-Transformed Measures</i>  |                                                                            |
| at_lag, sale_lag                             | One-year lagged values of total assets and sales                           |
| asset_g, sales_g                             | Growth rates: $\frac{X_t - X_{t-1}}{X_{t-1}}$ for assets and sales         |
| ln_at, ln_sale, ln_emp                       | Logarithms of firm size, sales, and employment                             |
| <i>Deal Indicators (CRSP/SDC)</i>            |                                                                            |
| deal_acq, deal_tgt                           | Binary indicators for acquirer and target firm-years                       |
| linkdt, linkenddt, link_span_days            | Link coverage period between CRSP and Compustat                            |